Software Interface vs Implementation Analogies

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Java Frameworks and APIs

Domain	Standard Interface / Spec	Implementation / Tools	Notes
Java Logging	SLF4J, java.util.logging	Logback, Log4j2, tinylog	Bindings enab
Java Persistence	JPA (javax.persistence)	Hibernate, EclipseLink, DataNucleus	Interface-drive
Dependency Injection	JSR-330 (`@Inject`)	Spring, Guice, Jakarta CDI	Spring adds A
Messaging / Events	JMS (Java Message Service)	ActiveMQ, RabbitMQ, IBM MQ	Pluggable tra
JAX-RS (Java API for REST)	JAX-RS Spec	RESTEasy, Jersey, Apache CXF	Interface spec
JSON Binding	JSR-367 (JSON-B)	Jackson, Gson, Yasson	All parse JSO
Testing	JUnit API	JUnit 5, TestNG, Spock	Test abstracti
Reactive Programming	Reactive Streams Spec	Reactor, RxJava, Akka Streams	API spec vs in

API & Web

Domain	Standard Interface / Spec	Implementation / Tools	Notes
REST API Design	REST principles / HTTP spec	Spring MVC, JAX-RS, Micronaut, Quarkus	Same verbs,
API Gateway	Reverse proxy pattern	Spring Cloud Gateway, NGINX, Kong, Istio Gateway	Same concep
API Documentation	OpenAPI (Swagger) Specification	Swagger UI, Redoc, Springdoc	Spec vs prese
GraphQL APIs	GraphQL Spec	Apollo, GraphQL Java, Netflix DGS	Spec vs serve
Configuration Management	Spring Boot Config Spec	Spring Cloud Config, Consul Config	Spec for confi

Containers & Orchestration

Domain	Standard Interface / Spec	Implementation / Tools	Notes
Docker / Containers	OCI (Open Container Initiative)	Docker, Podman, containerd, cri-o	Interchangeal
Container Orchestration	Kubernetes API	kubectl, Lens, ArgoCD, Helm, Kustomize	Helm = zsh of
Cloud Abstraction	Kubernetes API spec	AWS EKS, GKE, AKS, K3s	Same interfac

Infrastructure & DevOps

Domain	Standard Interface / Spec	Implementation / Tools	Notes
Build Tools	Build lifecycle (compile, test)	Maven, Gradle	Same phases
Infrastructure as Code	HashiCorp Configuration Language (HCL)	Terraform, Pulumi	Terraform = C
Tracing / Monitoring	OpenTelemetry API	Jaeger, Zipkin, Prometheus with exporters	Spec vs actua
Security	Spring Security interface	OAuth2, JWT, LDAP, SAML implementations	Same SPI, dif
Service Discovery	Service Registry concept	Eureka, Consul, Zookeeper	All expose se
Caching / Data Store	Redis protocol (RESP)	Redis OSS, AWS Elasticache, Azure Redis Cache	Protocol spec